

Problem set 2 solution

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1 Problems Set 2

Problem 1.1 (25pt). Let a_n and b_n be a sequence of real numbers.

1. Assume that $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ exist. Show that

$$\lim_{n \rightarrow \infty} (a_n b_n) = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right).$$

2. Give an example in which $\lim_{n \rightarrow \infty} (a_n b_n)$ exists but neither $\lim_{n \rightarrow \infty} a_n$ nor $\lim_{n \rightarrow \infty} b_n$ exists.

Problem 1.2 (25pt). Find the limit for the following sequence if it exists. Or show that the limit doesn't exist.

1. $a_n = \frac{n^2}{n+1} - \frac{n^2+1}{n}$

2. $a_n = \frac{\sin(n)}{n}$

- 3.

$$a_n = \underbrace{\frac{n^2}{\sqrt{n^6+1}} + \frac{n^2}{\sqrt{n^6+2}} + \frac{n^2}{\sqrt{n^6+3}} + \cdots + \frac{n^2}{\sqrt{n^6+n}}}_{n \text{ terms}}$$

Problem 1.3 (15pt). Let a_n be a sequence of real numbers and L be a real number. Show that the following two statements are equivalent. One holds if and only if the other does.

- There exists a subsequence a_{n_k} converging to L .
- For any $\epsilon > 0$, there exist infinite any a_n in $(L - \epsilon, L + \epsilon)$.

Proof. (Forward direction) $\implies$ Assume that there exists subsequence a_{n_k} such that $a_{n_k} \rightarrow L$. Then for any $\epsilon > 0$, there exists positive integer K such that for all $k \geq K$

$$|a_{n_k} - L| < \epsilon$$

then $L - \epsilon < a_{n_k} < L + \epsilon$ for all $k \geq K$. There are infinitely many a_n such that $a_n \in (L - \epsilon, L + \epsilon)$ for all $\epsilon > 0$ $\square$

Proof. (Backward direction) $\Leftarrow$ Assume that for all $\epsilon > 0$ there exists infinite a_n such that $a_n \in (L - \epsilon, L + \epsilon)$ then for each $k \in \mathbb{N}$ there are infinitely many i such that $a_i \in (L - \frac{1}{k}, L + \frac{1}{k})$ then for each k , we choose n_k such that $n_k > n_{k-1}$ and $|a_{n_k} - L| \leq \frac{1}{k}$ for each $\epsilon > 0$ there exists some integer K such that $K > \frac{1}{\epsilon}$ (Archimedean Property) so that for all $k \geq K$

$$|a_{n_k} - L| \leq \frac{1}{k} \leq \frac{1}{K} < \epsilon$$

Hence, the subsequence a_{n_k} converges to L $\square$

Problem 1.4 (15pt). Where possible find a subsequence that is monotone and a subsequence that is convergent for the following sequences.

1. $a_n = \sin(n\pi/8)$

2. $a_n = (-1)^n n$